Theorem 11: upper bound for the cut distance

Proof overview of the Lean 4/Mathlib formalisation

Folder `GsimOT/Theorem11_CutUpperBound`

Paper. J. Breitung, T. Gervens, M. Grohe. *Graph Similarity Meets Optimal Transport.*
File `Graph_Similarity_Meets_Optimal_Transport.pdf` in the repository; statement, equation, page and line numbers refer to this PDF. `docs/NUMBERING.md` lists the Lean files and declarations of all numbered statements.

1 Statement

Paper (Theorem 11, p. 10, l. 333–337; proof overview pp. 10–11, l. 338–380; proof in Appendix E.2, pp. 32–44). There is a universal constant $C > 0$ such that for all graphs G, H of the same order, with $\delta_\square = \delta_\square(G, H)$ and $\delta_\square^{\mathrm{OT}} = \delta_\square^{\mathrm{OT}}(G, H)$,

$$\delta_\square^{\mathrm{OT}} \le \delta_\square \le C\,\delta_\square^{\mathrm{OT}} \ln \frac{e}{\delta_\square^{\mathrm{OT}}} \ln\Big(e \ln \frac{e}{\delta_\square^{\mathrm{OT}}}\Big),$$

the right-hand side being 0 if $\delta_\square^{\mathrm{OT}} = 0$. Consequently $\delta_\square = O((\delta_\square^{\mathrm{OT}})^{1-\varepsilon})$ for every $\varepsilon > 0$; Theorem 2 (p. 3, l. 118–120) states this as $\delta_\square \le C_\varepsilon(\delta_\square^{\mathrm{OT}})^{1-\varepsilon}$, and E.2.3 (p. 44, l. 1377–1379) as $\delta_\square \le C_\alpha(\delta_\square^{\mathrm{OT}})^{\alpha}$ for every $\alpha < 1$.

Lean. The paper cites the thin-matching theorem of Haqi and Oveis Gharan (Theorem 35 = [HOG26, Thm. 1.4]); it is the hypothesis `hThin : ThinPermutationTheorem`.

- `GsimOT.theorem11_of_thin hThin`: $\exists C > 0$ such that for all finite types V, W, graphs G on V, H on W with $|V| = |W|$: $\delta_\square^{\mathrm{OT}} \le \delta_\square$ and $\delta_\square \le C f \ln(e/f) \ln(e \ln(e/f))$ with $f = \delta_\square^{\mathrm{OT}}$. (In Lean $\ln 0 = 0$ and $e/0 = 0$, so the right-hand side is 0 for $f = 0$, as in the paper.)
- `GsimOT.theorem11_power_lt_one_of_thin hThin`: for every $\alpha < 1$ there is $C_\alpha > 0$ with $\delta_\square \le C_\alpha f^\alpha$ (the statement of E.2.3; `theorem11_power_of_thin` is the special case $0 < \alpha < 1$).
- `GsimOT.theorem2_cut_of_thin hThin`: for every $\varepsilon > 0$ there is $C_\varepsilon > 0$ with $\delta_\square \le C_\varepsilon f^{1-\varepsilon}$ (Theorem 2, cut part).
- `theorem11_of_rackeShah`, `theorem11_power_lt_one_of_rackeShah`, `theorem2_cut_of_rackeShah` (`FromThinMatching.lean`): the same, assuming only the tree-cut sparsifier theorem [RS14], since Theorem 35 is derived from the separate library `ThinMatching/`.

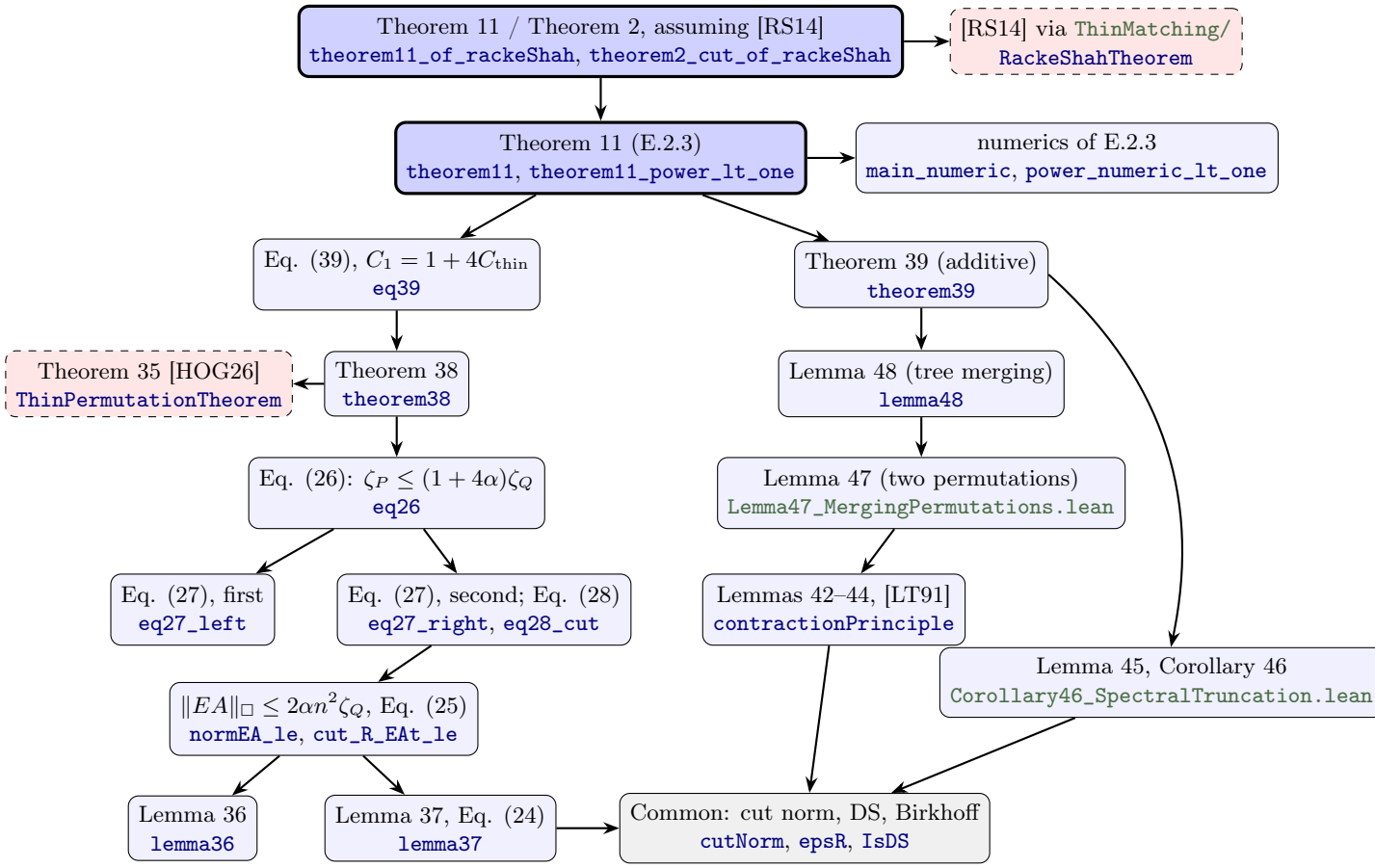
These are exactly the paper's statements (with the paper's constant $C = \max\{2, 39C_1, 113\}$, $C_1 = 1+4C_{\mathrm{thin}}$).

2 Where the proof lives

File	Content
`Definitions.lean`	thinness Eq. (23) (`IsThin`), Theorem 35 as a hypothesis (`ThinPermutationTheorem`), the contraction principle statement, support functions, d_n
`Theorem38_ThinRounding.lean`	Appendix E.2.1: Lemma 36 (`lemma36`), Lemma 37 (`lemma37`), Eqs. (24)–(28), Eq. (26) (`eq26`), Theorem 38 (`theorem38`)
`Corollary46_SpectralTruncation.lean`, `Lemma47_MergingPermutations.lean`, `ContractionPrinciple.lean`, `Theorem39_AdditiveRounding.lean`	Appendix E.2.2: the additive rounding, Theorem 39 (`theorem39`)
`Theorem11.lean`	Appendix E.2.3: graph facts, numerics, Eq. (39) (`eq39`), Theorem 11 (`theorem11`), the power bounds and Theorem 2
`FromThinMatching.lean`	Theorem 35 from `ThinMatching/` and the `_of_rackeShah` versions

The paper's ζ_Q (Eq. (21)) is `epsR (adjMat G) (adjMat H) Q`, and $\zeta_Q = n^{-2}C_\square(Q)$ (`epsR_adjMat`).

3 Proof layout



Left: the multiplicative branch (Appendix E.2.1); right: the additive branch (Appendix E.2.2); top: the combination (Appendix E.2.3). Dashed boxes are cited results, which are explicit hypotheses.

4 The multiplicative bound (Appendix E.2.1, pp. 32–35)

Throughout Appendix E.2, $V = W = [n]$ with $n \ge 2$, A, B are the adjacency matrices, and ζ_Q is defined by Eq. (21) (sets) = Eq. (22) ($[0,1]$-valued functions), p. 32. In Lean V, W are arbitrary finite types of equal size; testing on sets is `abs_cut_le_epsR`.

1. **Thinness, Eq. (23)** (p. 32, l. 1014–1015): P is α-thin w.r.t. Q if for all X, Y the mass of P on the cut $(X \times \overline{Y}) \cup (\overline{X} \times Y)$ is at most α times that of Q. `IsThin` is this, written with $U = U_V \sqcup U_W$ as in the proof overview (p. 10, l. 350–353).

2. **Theorem 35** (p. 32, l. 1021–1022) is the hypothesis `ThinPermutationTheorem`: some P is $C_{\text{thin}} \ln(2n) \ln\ln(4n)$-thin.

3. **Lemma 36** (p. 33, l. 1026–1028): $\sum_i |x_i - y_{\pi(i)}| \le \alpha \sum_{ij} Q_{ij} |x_i - y_j|$. As in the paper: Eq. (23) for the level sets X_s, Y_s gives the pointwise inequality (`hpt` in `lemma36`, with `levelInd_apply`), and $\int |1[x > s] - 1[y > s]|\, ds = |x - y|$ (`integral_levelInd`); integrate.

4. **Lemma 37** (p. 33, l. 1038–1039; proof pp. 33–34): $n^{-2} \sum_{ij} Q_{ij} |x_i - (Q^\top x)_j| \le 4\zeta_Q$ for $x = A\mathbf{1}_T$ (`lemma37`; the paper allows $y \in [0,1]^n$, only $y = \mathbf{1}_Y$ is used). Steps: $F_{ij} = x_i - r_j$, Eq. (24) $n^{-2} \sum Q_{ij} |F_{ij}| \le 2\zeta_Q$ via $S_\pm$, then $x_i - (Q^\top x)_j = F_{ij} - \sum_{i'} Q_{i'j} F_{i'j}$ and the triangle inequality.

5. **Eq. (25)** (p. 34, l. 1066–1072): $\|RK\|_\square \le \|K\|_\square$, used in the form `abs_sum_mul_le_of_cut`.

6. **Theorem 38** (p. 34, l. 1074–1075; proof pp. 34–35): $\zeta_P \le (1+4\alpha)\zeta_Q$ for some P (`theorem38`). Its proof:

   (a) $\zeta_P = n^{-2} \|A - PBP^\top\|_\square$ (p. 34, l. 1080–1084): `epsR_permMat_le` (direction $\le$).

(b) split $A - PBP^\top = (A - RAR^\top) + P(Q^\top AQ - B)P^\top$, $R = PQ^\top$, i.e. $R(v,k) = Q(k,\pi v)$ (p. 35, l. 1085–1089): `hsplit` in `eq26`; conjugation invariance by testing on $\pi(S) \times \pi(T)$.

(c) Eq. (27), first bound $n^{-2}\|Q^\top AQ - B\|_\square \le \zeta_Q$ (p. 35, l. 1091–1095): `eq27_left`.

(d) $f = EA\mathbf{1}_Y$ with $E = I - R$; Lemma 36, Lemma 37, $\sum_i f_i = 0$ give $n^{-2}\|EA\|_\square \le 2\alpha\zeta_Q$ (l. 1096–1105): `normEA_le`, with `abs_sum_le_half_of_sum_eq_zero`.

(e) Eq. (28) $A - RAR^\top = EA + R(EA)^\top$ (symmetry of A): `eq28_cut`; Eq. (25) and transposition: `cut_R_EAt_le`; together the second bound of Eq. (27): `eq27_right`.

(f) Eq. (26) for every α-thin P: `eq26`; with Theorem 35: `theorem38`.

5 The additive bound (Theorem 39)

5.1 Statement

Paper (Theorem 39, p. 36, l. 1112–1114, Appendix E.2.2). Let $d_n := \lceil 2\log n \rceil$ (log is the binary logarithm, $n \ge 2$). For every $Q \in \mathrm{DS}(n)$ there is a permutation matrix P with

$$\zeta_P \le \zeta_Q + 36\sqrt{d_n}\, n^{-1/6}.$$

Lean. `GsimOT.theorem39` (file `Theorem39_AdditiveRounding.lean`): for finite types V, W with $|V| = |W| = n \ge 2$, symmetric matrices $A \in \mathbb{R}^{V\times V}$, $B \in \mathbb{R}^{W\times W}$ with entries in $[0,1]$ and $Q \in \mathrm{DS}(V,W)$ there is a bijection $\pi\colon V \to W$ with

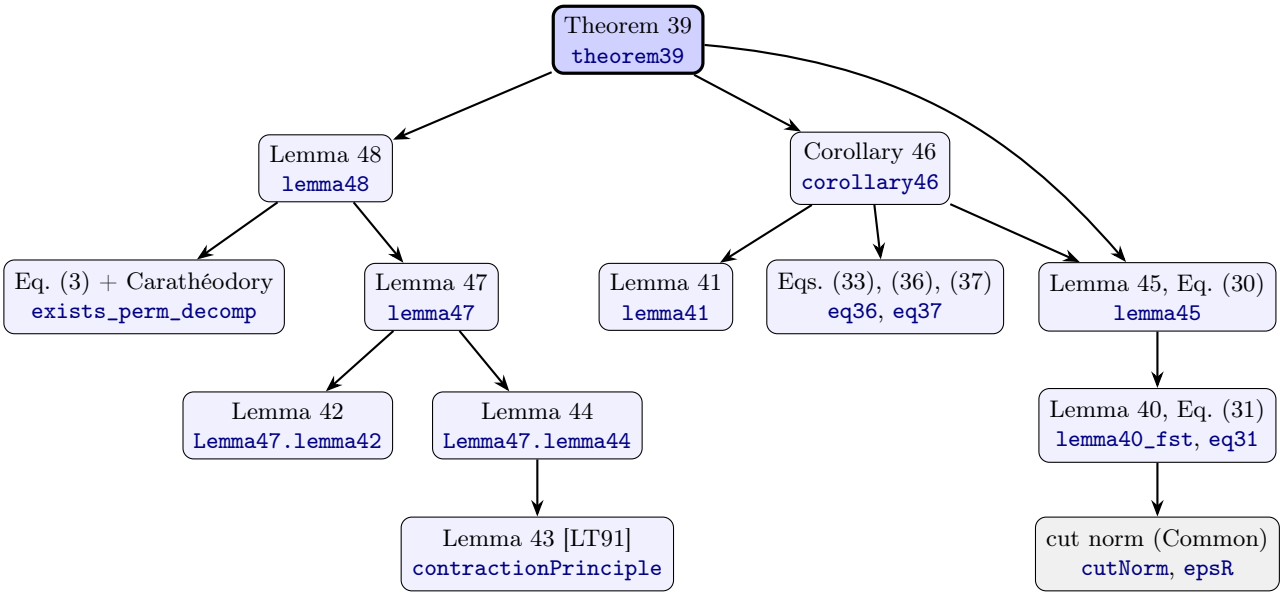
$$\texttt{epsR A B (permMat pi)} \le \texttt{epsR A B Q} + 36\sqrt{d_n}\, n^{-1/6}, \qquad d_n = \texttt{dn n} = \lceil \log_2(n^2) \rceil.$$

Here `epsR A B R` is $\zeta_R(A,B)$, the weighted cut norm of the discrepancy kernel on (Ω_R, μ_R); for adjacency matrices it is ζ_R (Eq. (21)). The hypothesis **hCP : ContractionPrinciple** (Lemma 43, [LT91]) is discharged by `GsimOT.contractionPrinciple`. The statement is the paper's (for adjacency matrices of graphs of the same order), slightly more general (any symmetric $[0,1]$-matrices).

5.2 Files

File	Content
`Definitions.lean`	`radAvg`, `ContractionPrinciple` (statement of Lemma 43), `hSupp` (h_Q, Eq. (33)), `rhoSupp` (ρ, Eq. (34)), `dn`
`ContractionPrinciple.lean`	Lemma 43 proved: `contractionPrinciple` (Talagrand's comparison argument)
`Corollary46_SpectralTruncation.lean`	Lemma 40, spectral truncation and Eq. (31), Lemma 45 (Eq. (30)), zonotopes and Eqs. (32), (33), (36), (37), Lemma 41, Corollary 46
`Lemma47_MergingPermutations.lean`	Lemma 42 (`Lemma47.lemma42`), Lemma 44, the blocks, Eq. (38), Lemma 47
`Theorem39_AdditiveRounding.lean`	Eq. (35), Eq. (3) with Carathéodory, Lemma 48, Theorem 39

5.3 Proof layout



Lemma 48 also uses Eq. (35) (`rhoSupp_convex`); all results in this part are proved, the only external input (Lemma 43, cited in the paper) is proved in `ContractionPrinciple.lean`.

5.4 The proof step by step

Notation (p. 36). The paper works in $\mathcal{L}^2(n)$ (inner product $\frac{1}{n}\sum_i x_i y_i$) with averaging operators. Lean uses the standard inner product and rescales: the paper's eigenvalues are λ_r/n, its eigenfunctions $\sqrt{n}\,e_r$, and $\|K\|_{\mathcal{L}^2(n)} \le t$ becomes `OpBound K (t * n)` ($|\sum_{i,k} f_i K_{ik} g_k| \le tn \cdot \frac{\|f\|^2+\|g\|^2}{2}$, equivalent by scaling). The couplings $\mu_Q(i,j) = Q_{ij}/n$ and $\zeta_Q(A',B') = \|D_{A',B'}\|_{\square,\mu_Q}$ are built into `epsR`.

Lemma 40 (Lifted Kernels, p. 36). `lemma40_fst`, `lemma40_snd`. As in the paper, for $S,T \subseteq \Omega_Q$ the marginal weights $s_i = \sum_j Q_{ij}\mathbf{1}_S(i,j) \in [0,1]$ have $\|s\| \le 1$ (`weight_sq_le`), and the cut value is $\langle s, K\sigma\rangle$, bounded by the operator norm.

Lemma 41 (Close Support Functions, p. 37). `lemma41`, for non-empty compact Z, Z' in any real inner product space (without the convexity hypothesis), $\|T\| \le \tau$, with $h_Z =$ `supportFn` and $\max_{u,u'\in Z} |\langle u, Tu'\rangle| =$ `bilMax`. The paper's steps: $|h_Z(w) - h_{Z'}(w)| \le \eta\|w\|$ (`abs_supportFn_sub_le`, by homogeneity `supportFn_smul`); $m_T(Z,Z) \le m_T(Z',Z) + M\eta\tau$ via $h_Z(Tu')$, and $m_T(Z',Z) \le m_T(Z',Z') + M\eta\tau$ via $h_{Z'}(T^*u)$ with the adjoint (`lemma41_step`); the same for $-T$ gives the absolute values (`abs_inner_le_bilMax_add`).

Lemma 42 (symmetrisation, pp. 37–38). `Lemma47.lemma42`: independent $X_a \in \{0,1\}$ with $\mathbb{E}X_a = \theta_a$ (product weights `bern`), bounded d_a; independent copy X' and Jensen, symmetry of $X_a - X'_a$ (coordinate swap `swapAt`), triangle inequality — as in the paper.

Lemma 43 (Talagrand Contraction Lemma, p. 38). Cited in the paper; stated as `ContractionPrinciple` (for non-empty bounded S) and proved by `contractionPrinciple`: the two-point lemma, the comparison theorem without absolute values (one coordinate at a time), then absolute values and the factor 2 by adding 0 to S and using $\sigma \mapsto -\sigma$.

Lemma 44 (p. 38, proof pp. 38–39, Eq. (29)). `Lemma47.lemma44`: for non-empty blocks B_a and $\|u_i\| \le M$, $\mathbb{E}_\sigma \sup_{w\in B_H} |\sum_a \sigma_a \sum_{i\in B_a} \langle w, u_i\rangle_+| \le 2M(\sum_a \ell_a^2)^{1/2}$. As in the paper: $\sum_{i\in B_a}\langle w,u_i\rangle_+ = \mathbb{E}_{I_a}(\ell_a\langle w, u_{I_a}\rangle)_+$ with uniform independent $I_a \in B_a$ (`sum_piFinset_comp`, over `Fintype.piFinset B`); Jensen; for fixed I Lemma 43 with $\gamma(s) = s_+$; $\sup_w |\langle w, g_\sigma\rangle| \le \|g_\sigma\|$; $\mathbb{E}\|g_\sigma\| \le (\mathbb{E}\|g_\sigma\|^2)^{1/2}$ (`radAvg_norm_le`).

Lemma 45 (spectral approximation, p. 39, proof pp. 39–40, Eqs. (30), (31)). `lemma45`. Spectral decomposition and Parseval (`spectral_entry`, `parseval_row`); truncations A_t, B_t (`specTrunc`) with $\|A - A_t\|_{\mathcal{L}^2(n)} \le t$ (`opBound_sub_specTrunc`); Eq. (31), $|\zeta_Q(A,B) - \zeta_Q(A_t,B_t)| \le 2t$, from $D_{A,B} = D_{A_t,B_t} + (A-A_t)^{(1)} - (B-B_t)^{(2)}$ and Lemma 40 (`eq31`); the vectors z_{ij} (`specVec`, $\|z_{ij}\| \le \sqrt{2}$ by Parseval, `norm_specVec_le`) and $T_t = \mathrm{diag}(\lambda_r^{-1}, -\nu_s^{-1})$ (`diagOp`, `specDiag`, $\|T_t\| \le 1/t$ by `norm_diagOp_le`) with $\langle z_{ij}, T_t z_{k\ell}\rangle = (A_t)_{ik} - (B_t)_{j\ell}$ (`diagForm_specVec`, `inner_diagOp`). The maximum in Eq. (30) is the cut norm of `zonKernel`. Lean takes $H = \mathbb{R}^{V \sqcup W}$ (the coordinates outside $I_A \sqcup I_B$ are zero), so the case $I_A = I_B = \emptyset$ is not separate.

Zonotopes and ρ (pp. 40–41, Eqs. (32)–(37)). Z_Q is the image of `cube` under `zonLin v Q` (Eq. (32)), compact, convex, in the ball of radius M (`isCompact_zonotope`, `convex_zonotope`, `norm_zonLin_le`); Eq. (33), $h_{Z_Q} = h_Q$ (`supportFn_zonotope`, maximiser $a_{ij} = \mathbf{1}[\langle w, z_{ij}\rangle > 0]$); Eq. (34) is `rhoSupp`; Eq. (35) (`rhoSupp_triangle`, `rhoSupp_convex`); Eq. (36) as an identity (`eq36`: bilinear expansion `eq36_expand`, maximum at indicator vectors); Eq. (37) (`eq37`).

Corollary 46 (p. 41). `corollary46`: for v, T satisfying Eq. (30) and all $P, Q \in \mathrm{DS}(n)$, $\zeta_P \le \zeta_Q + 4t + \frac{2\sqrt{2}}{t}\rho(P,Q)$: Lemma 41 with $M = \sqrt{2}$, $\eta = \rho(P,Q)$, $\tau = 1/t$, and Eq. (37) for P and Q — exactly the paper's chain.

Lemma 47 (Merging Two Permutations, pp. 41–43). `lemma47` (statement, $20Mn^{-1/3}$) from `lemma47_sharp` ($18Mn^{-1/3}$, what the proof gives on p. 43, l. 1333–1334). Steps: $\tau = \pi_0^{-1}\pi_1$ and $u_i = z_{i,\pi_0(i)}$ (the w.l.o.g. $P_0 = I$); $L = \lceil n^{1/3} \rceil$; blocks with properties (i), (ii) (`exists_blocks`, from a maximal L-separated set of cut points, `exists_sep_cover`); property (i) gives $\sum_a |B_a|^2 \le 2Ln$; $n(h_{T_X} - (1-\theta)h_{\mathrm{id}} - \theta h_\tau) = \sum_a (X_a - \theta)d_a$; Lemma 42 and twice Lemma 44 give some X with $\Delta(T_X) \le 8M\sqrt{2L/n} \le 16Mn^{-1/3}$ (Eq. (38), `exists_good_choice`); repair of T_X outside the at most n/L block boundaries (`exists_equiv_eq_off`), each changed row costing $\le 2M/n$ (`phi_sub_le`).

Lemma 48 (p. 43). `lemma48`: Birkhoff (Eq. (3)) and Carathéodory give $Q = \sum_{s \le N} \lambda_s P_s$ with $\lambda_s > 0$, $N \le n^2$ (`exists_perm_decomp`, via Mathlib's Birkhoff theorem and `eq_pos_convex_span_of_mem_convexHull`; the bound $N \le n^2$ from affine independence in $\mathbb{R}^{n \times n}$ with constant total mass); a binary tree of height $\le d_n$ (`merge_aux`: induction on the height k, splitting the index set in halves; `combo` is Q_ν); at each node Lemma 47, the triangle inequality and Eq. (35) give $\rho(P_\nu, Q_\nu) \le \eta_0 + (k-1)\eta_0$.

Proof of Theorem 39 (p. 43). `theorem39`: $t = \sqrt{20 d_n}\, n^{-1/6}$, Lemma 45 for this t, Lemma 48 ($\rho(P,Q) \le 20\sqrt{2}\, d_n n^{-1/3}$), Corollary 46: $\zeta_P \le \zeta_Q + 4t + 80 d_n n^{-1/3}/t = \zeta_Q + 8t$ and $8\sqrt{20} < 36$ (`hs20`: $\sqrt{20} \le 9/2$).

5.5 Notes on the formalisation (additive part)

- Standard inner product instead of $\mathcal{L}^2(n)$; $H = \mathbb{R}^{V \sqcup W}$ instead of $\mathbb{R}^{I_A \sqcup I_B}$.
- Lemma 41 is stated without the convexity hypothesis and Lemma 44 without the disjointness hypothesis; the paper's arguments prove these more general forms.
- Lemma 47: blocks from a maximal separated set instead of the cycle decomposition (only properties (i), (ii) are used); Lemma 44 is applied to all blocks with vectors $X_a u_i$ instead of only the blocks with $X_a = 1$; the proof gives the constant $18M$, as in the paper's proof.
- Lemma 48: the binary tree halves the index set at every node instead of padding with zero-weight leaves.
- Expectations are finite weighted sums (`radAvg`, `bern`, uniform averages over a `piFinset`).

The pseudometric ρ is formalised by its definition via support functions, Eq. (34) (`rhoSupp`); for the compact convex zonotopes Z_Q this is the Hausdorff distance of the proof overview (p. 11, l. 368–369).

5.6 How to check

- `lake build GsimOT.Theorem11_CutUpperBound.Theorem39_AdditiveRounding`
- `#print axioms GsimOT.theorem39`: only `propext`, `Classical.choice`, `Quot.sound`.

6 Combining both bounds: the proof of Theorem 11 (Appendix E.2.3, p. 44)

1. *Lower bound* (l. 1363): permutation matrices are doubly stochastic: `deltaOTCut_le_deltaCut`.
2. *Optimal Q*: "let $Q \in \mathrm{DS}(n)$ be optimal, so that $\zeta_Q = \varepsilon$": attained by compactness of $\mathrm{DS}(V, W)$ and continuity of $C_\square$ (`isCompact_DS`, `continuous_costCut`, `exists_costCut_eq_dOTCut`).
3. *Eq. (39)* (l. 1364–1366): $\ln(2n)\ln\ln(4n) \geq \ln 4 \cdot \ln\ln 8 > 1$ for $n \geq 2$ (`one_lt_log_four_mul_log_log_eight`, `one_le_log_mul_log_log`), so Theorem 38 gives $\delta_\square \leq C_1 \ln(2n)\ln\ln(4n)\,\varepsilon$ with $C_1 = 1 + 4C_{\mathrm{thin}}$: `eq39`.
4. *Cases* (all in `main_numeric`): $\varepsilon = 0$ (from Eq. (39)); $\varepsilon \geq \frac{1}{2}$ ($\delta_\square \leq 1$, $C \geq 2$); $0 < \varepsilon < \frac{1}{2}$ with $L_\varepsilon = \ln(e/\varepsilon)$:
   - $n \leq \varepsilon^{-12}$: $\ln(2n) \leq 13 L_\varepsilon$ and $\ln\ln(4n) \leq \ln(14 L_\varepsilon) \leq 3\ln(e L_\varepsilon)$ using $\ln 14 < 3$ (`log_fourteen_lt_three`), hence $C = 39 C_1$;
   - $n > \varepsilon^{-12}$: $d_n \leq 3 \log n$ (`dn_mul_log_two_le`) and the error term of Theorem 39 is $< 112\varepsilon$ (`error_term_le`), hence $\delta_\square \leq 113\varepsilon$.

   So $C = \max\{2, 39C_1, 113\}$ (`theorem11`).
5. *Power bound* (l. 1377–1379): $\varepsilon^{1-\alpha}\ln(e/\varepsilon)\ln(e\ln(e/\varepsilon))$ is bounded on $(0, 1]$: `power_numeric` ($0 < \alpha < 1$, with constant $(1 + \frac{2}{1-\alpha})^2$) and `power_numeric_lt_one` (every $\alpha < 1$; for $\alpha \leq \frac{1}{2}$ via $\alpha = \frac{1}{2}$ and $f^{1/2} \leq f^\alpha$). Results: `theorem11_power_lt_one`, `theorem2_cut_of_thin` ($\alpha = 1 - \varepsilon$).

7 Assumptions and trusted base

- **Theorem 35** [HOG26, Thm. 1.4] is the hypothesis `ThinPermutationTheorem` of the `_of_thin` theorems; it is derived from [RS14] (`RackeShahTheorem`, cited in [HOG26]) in the `ThinMatching/` library, so the `_of_rackeShah` theorems assume only [RS14].
- The contraction principle [LT91] (Lemma 43) is a hypothesis of `theorem11`, but it is proved (`contractionPrinciple`), so the `_of_thin` theorems do not assume it.
- Only the standard axioms `propext`, `Classical.choice`, `Quot.sound` are used.

8 Notes on the formalisation

- Lemma 37 is proved for $y = \mathbf{1}_T$ only (the case used).
- The Lean hypothesis gives thinness with exactly $\alpha = C_{\mathrm{thin}}\ln(2n)\ln\ln(4n)$, so the monotonicity step of the proof of Theorem 38 ("α'-thin, $\alpha' \leq \alpha$, hence α-thin") is not needed.
- $n = 1$ (excluded by "$n \geq 2$" in Appendix E.2) is handled separately: $\delta_\square = 0$ (`deltaCut_le_zero_of_card_eq_one`, `rhs_nonneg`).
- For $n > \varepsilon^{-12}$ the paper uses that $x \mapsto x^{-1/6}\sqrt{\ln x}$ is decreasing and $\varepsilon\sqrt{\ln(1/\varepsilon)} \leq (2e)^{-1/2}$; Lean uses $m = n^{1/6} > \varepsilon^{-2}$ and $\ln m \leq m/e$ (the same inequality), with the same constant 112.

9 How to check this result

```
lake build GsimOT.Theorem11_CutUpperBound.Theorem11 GsimOT.Theorem11_CutUpperBound.FromThinMatching
lake env lean scripts/CheckAxioms.lean   # axioms of theorem11_of_thin, theorem11_of_rackeShah, ...
```

The correspondence table is `Theorem11_correspondence.pdf` in the same folder.